

01.WA.ENL1.5-BLO15-EMP1.5

Exterior wall
created on 17.5.2026

Thermal protection

$U = 2,59 \text{ W/(m}^2\text{K)}$

GEG 2020/24 Bestand*: $U < 0,24 \text{ W/(m}^2\text{K)}$



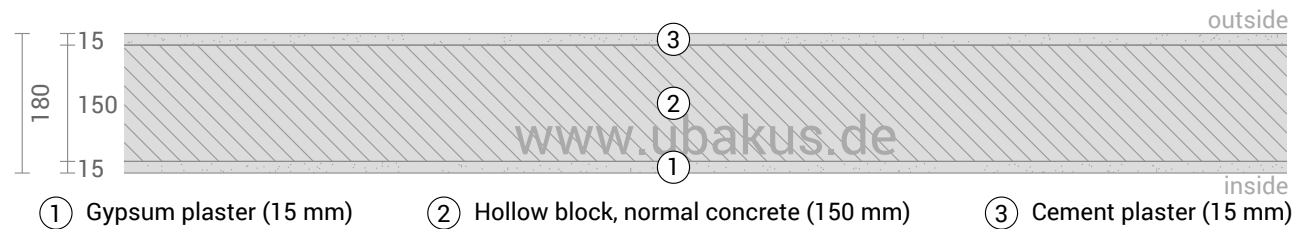
Moisture proofing

Surface temperature at inside too low!
Dries 10 days



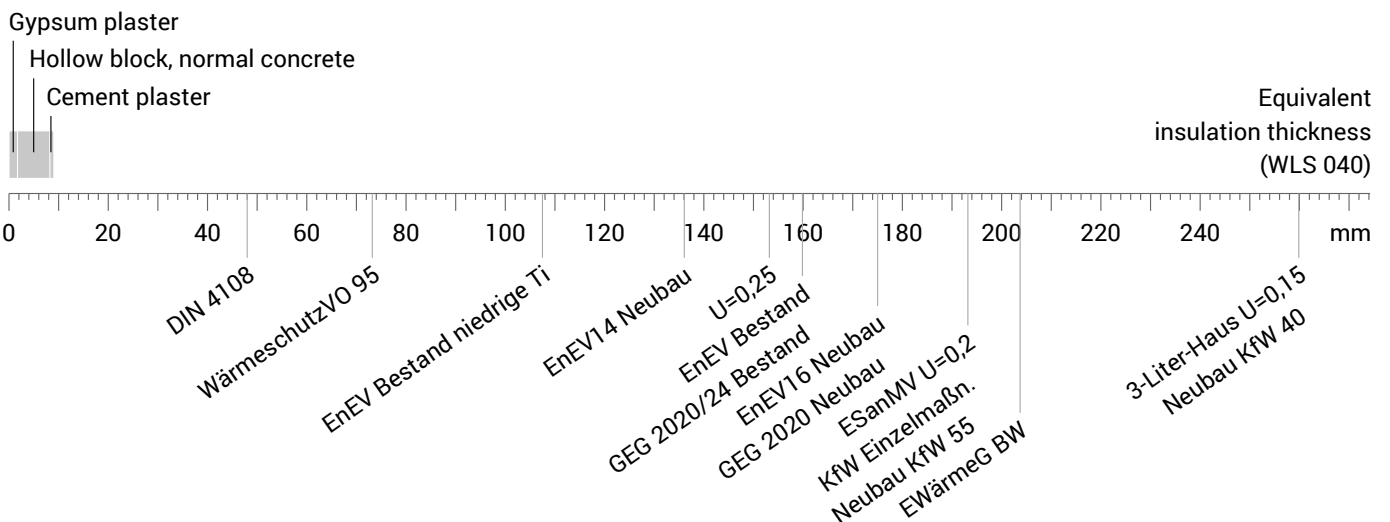
Heat protection

Temperature amplitude damping: 1,8
phase shift: 5,2 h
Thermal capacity inside: 77 kJ/m²K



Impact of each layer and comparison to reference values

For the following figure, the thermal resistances of the individual layers were converted in millimeters insulation. The scale refers to an insulation of thermal conductivity 0,040 W/mK.



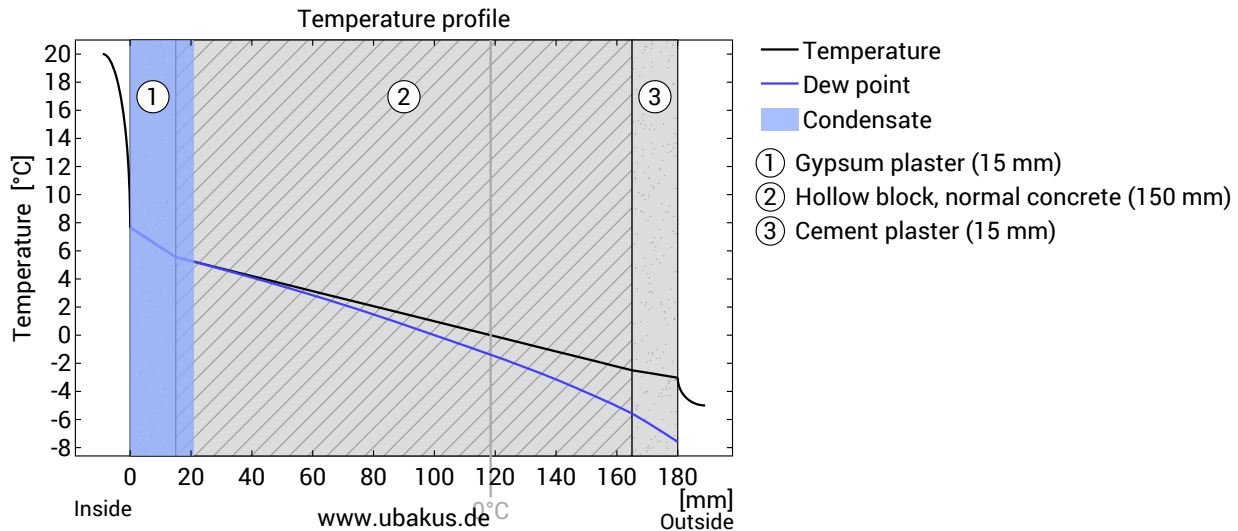
Inside air : 20,0°C / 50%
Outside air: -5,0°C / 80%
Surface temperature.: 7,7°C / -3,0°C

sd-value: 5,2 m

Thickness: 18,0 cm
Weight: 315 kg/m²
Heat capacity: 289 kJ/m²K

01.WA.ENL1.5-BLO15-EMP1.5, $U=2,59 \text{ W}/(\text{m}^2\text{K})$

Temperature profile



Temperature and dew-point temperature in the component. The dew-point indicates the temperature, at which water vapour condenses. As long as the temperature of the component is everywhere above the dew-point temperature, no condensation occurs. If the curves have contact, condensation occurs at the corresponding position.

Layers (from inside to outside)

#	Material	λ [W/mK]	R [m²K/W]	Temperatur [°C]		Weight [kg/m²]
				min	max	
	Thermal contact resistance*		0,130	7,7	20,0	
1	1,5 cm Gypsum plaster	0,350	0,043	5,5	7,7	15,0
2	15 cm Hollow block, normal concrete	0,920	0,163	-2,5	5,5	270,0
3	1,5 cm Cement plaster	1,400	0,011	-3,0	-2,5	30,0
	Thermal contact resistance*		0,040	-5,0	-3,0	
	18 cm Whole component		0,387			315,0

*Thermal contact resistances according to DIN 6946 for the U-value calculation. $R_{si}=0,25$ and $R_{se}=0,04$ according to DIN 4108-3 were used for moisture proofing and temperature profile.

Surface temperature inside (min / average / max): 7,7°C 7,7°C 7,7°C
Surface temperature outside (min / average / max): -3,0°C -3,0°C -3,0°C

01.WA.ENL1.5-BLO15-EMP1.5, $U=2,59 \text{ W}/(\text{m}^2\text{K})$

Moisture proofing

For the calculation of the amount of condensation water, the component was exposed to the following constant climate for 90 days: inside: 20°C und 50% Humidity; outside: -5°C und 80% Humidity. This climate complies with DIN 4108-3.

On the inside of this component, humidity will condense because the surface temperature (7,7°C) is below the dew point temperature (9,3°C). This will lead to mold formation in the long term.

You can prevent this by reducing the relative humidity of the room air or increasing the surface temperature by (additional) thermal insulation. The reduction in humidity is only recommended in exceptional cases or as a short-term measure.

#	Material	sd-value [m]	Condensate [kg/m ²] [Gew.-%]	Weight [kg/m ²]
1	1,5 cm Gypsum plaster	0,15	1,4	15,0
2	15 cm Hollow block, normal concrete	4,50	1,2	270,0
3	1,5 cm Cement plaster	0,53	-	30,0
	18 cm Whole component	5,18	1,4 (!)	315,0

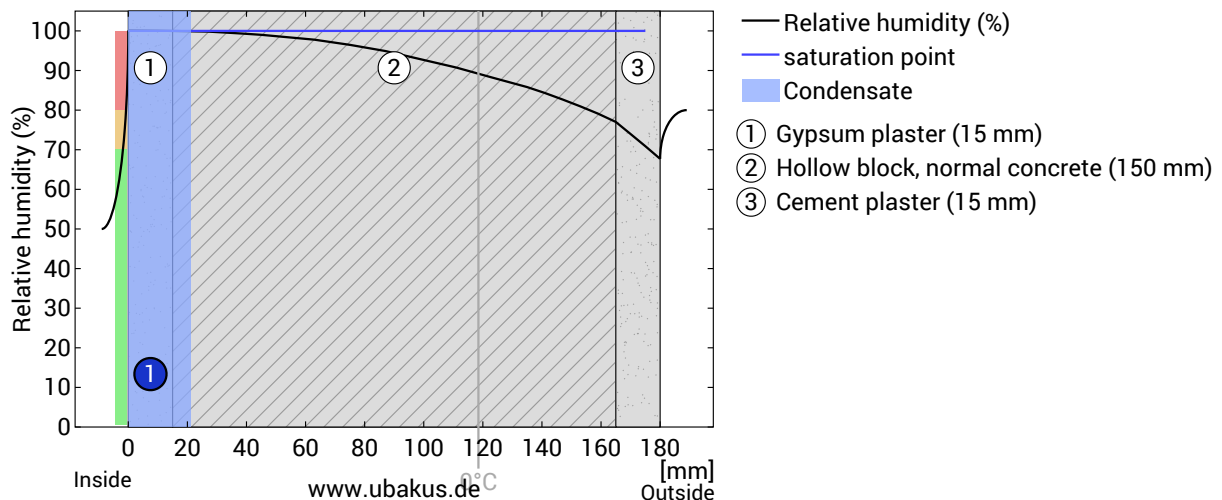
Condensation areas

- ① Condensate: Affected layers: Gypsum plaster, Hollow block, normal concrete

Humidity

The temperature of the inside surface is 7,7 °C leading to a relative humidity on the surface of 100%. Most kinds of moulds start to grow at relative air humidities of 80% or more. Mould grow is expected! To avoid mould formation, the surface temperature should be increased by (additional) insulation.

The following figure shows the relative humidity inside the component.

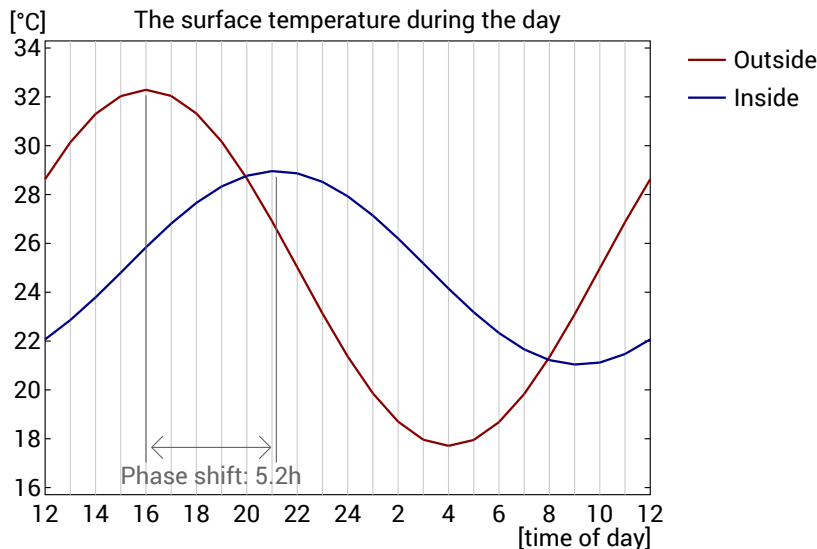
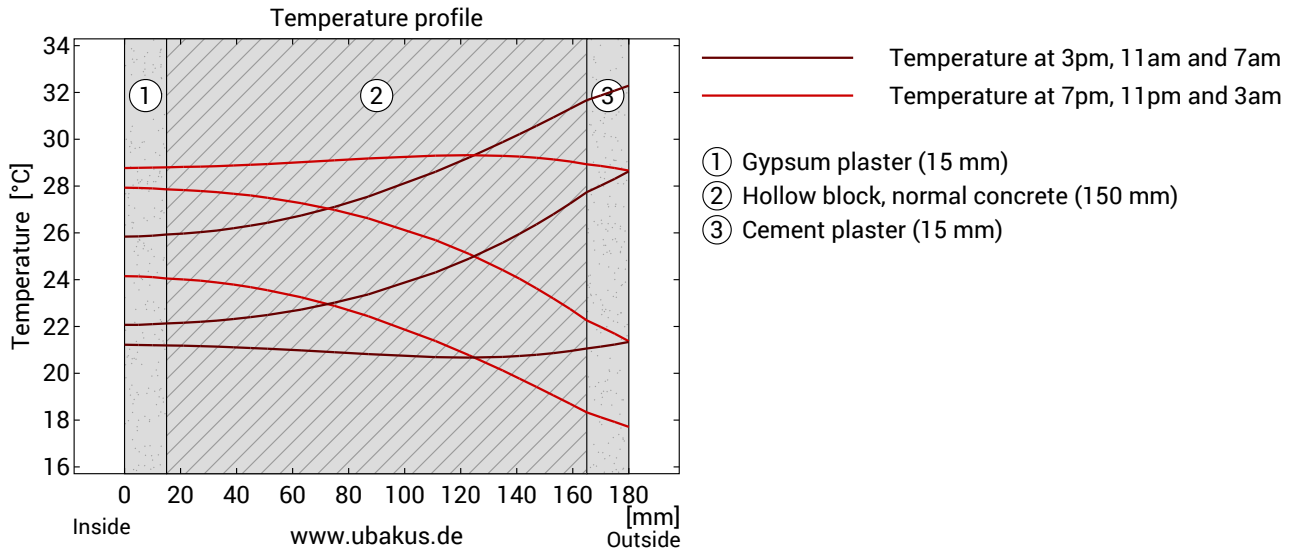


Notes: Calculation using the Ubakus 2D-FE method. Convection and the capillarity of the building materials were not considered. The drying time may take longer under unfavorable conditions (shading, damp / cool summers) than calculated here.

01.WA.ENL1.5-BLO15-EMP1.5, $U=2,59 \text{ W}/(\text{m}^2\text{K})$

Heat protection

The following results are properties of the tested component alone and do not make any statement about the heat protection of the entire room:



Top: Temperature profile within the component at different times. From top to bottom, brown lines: at 3 pm, 11 am and 7 am and red lines at 7 pm, 11 pm and 3 am.

Bottom: Temperature on the outer (red) and inner (blue) surface in the course of a day. The arrows indicate the location of the temperature maximum values . The maximum of the inner surface temperature should preferably occur during the second half of the night.

Phase shift*	5,2 h	Heat storage capacity (whole component):	289 kJ/m²K
Amplitude attenuation **	1,8	Thermal capacity of inner layers:	77 kJ/m²K
TAV ***	0,544		

* The phase shift is the time in hours after which the temperature peak of the afternoon reaches the component interior.

** The amplitude attenuation describes the attenuation of the temperature wave when passing through the component. A value of 10 means that the temperature on the outside varies 10x stronger than on the inside, e.g. outside 15-35 °C, inside 24-26 °C.

*** The temperature amplitude ratio TAV is the reciprocal of the attenuation: $TAV = 1 / \text{amplitude attenuation}$

Note: The heat protection of a room is influenced by several factors, but essentially by the direct solar radiation through windows and the total amount of heat storage capacity (including floor, interior walls and furniture). A single component usually has only a very small influence on the heat protection of the room.